

Magic2: a unified, self-configuring aligner for short- and long-read RNA-seq with comprehensive quality-control output

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Abstract

We present Magic2, a fast, self-configuring RNA-seq and DNA-seq aligner that operates identically on short (Illumina) and long (Nanopore, PacBio) reads without requiring the user to specify read type, adapter sequences, strandedness, or the number of threads. In a single pass, Magic2 produces alignments, splice-junction tables, gene-level expression counts, strand-specific coverage tracks, poly(A)/poly(T) tail calls, and a rich set of quality-control metrics — all directly comparable across runs and laboratories. Benchmarks on 35 public datasets (totalling ~70 Gb, including human, chimpanzee, and lemur reads aligned to the human T2T reference, as well as Nanopore and PacBio long-read datasets) show that Magic2 matches or exceeds leading aligners (STAR, HISAT2, Minimap2, Magic-BLAST) on alignment rate, concordant-pair fraction, mismatch rate, and transcript-level accuracy, while achieving competitive elapsed time and substantially lower CPU time than STAR. Magic2 is freely available at <https://github.com/NLM-DIR/Magic2>.

Introduction

High-throughput RNA sequencing is now the dominant approach for measuring gene expression, yet the computational chain that transforms raw reads into biological conclusions remains fragile and poorly standardised. Small differences in aligner choice, parameter settings, or post-processing steps routinely produce divergent results from identical input data (??). This fragility has driven the impractical recommendation that authors record every parameter down to the exact version of each software component — an implicit acknowledgement that reproducibility is not yet guaranteed by design.

A contributing source of variability is the proliferation of specialised tools: STAR (?) and HISAT2 (?) are optimised for short Illumina reads, while Minimap2 (?) targets long, error-prone reads. Users must master multiple programs with different parameter conventions and output formats, and the choice of default parameters can unintentionally bias benchmark comparisons. No single aligner currently handles the full spectrum of read technologies with consistent accuracy and without manual reconfiguration.

Magic2 addresses these limitations through three design principles. First, full automation: read length, paired-end geometry, adapter sequences, and sequencing platform are inferred from the data; internal parameters such as seed length and maximum intron size are derived from genome size. Second, single-pass richness: alignments, expression counts, splice-junction support, coverage tracks, poly(A) calls, and QC metrics are all produced in one run, eliminating downstream processing tools. Third, standardised QC: every run emits an identical set of quality metrics (fraction of reads mapping to CDS, introns, intergenic regions; strand-specificity score; concordant-pair rate; mismatch distribution), enabling large-scale, cross-laboratory comparisons of SRA datasets.

Results

Overview of Magic2

Magic2 accepts a reference genome (FASTA) and an optional annotation (GTF/GFF) to build an on-disk seed index. At run time, one or more SRA accession numbers or local FASTA/FASTQ files are supplied; no other configuration is required. Magic2 detects read length, adapter sequences, paired-end status, and strandedness automatically via a dry pass over the first reads. The same executable is used for Illumina short reads, PacBio CCS, and Nanopore long reads.

Internally, Magic2 follows a sort–merge paradigm rather than hash-table lookup: seed records are generated for reads and genome in large sequential blocks, sorted with a cache-efficient radix sort (optionally GPU-accelerated via CUDA Thrust), and joined by simultaneous linear traversal of both sorted lists. This design keeps the dominant computation in fast CPU cache rather than in random main-memory accesses, and scales naturally across NUMA architectures without user-supplied thread counts (see Supplementary Material for full details).

The pipeline is structured as concurrent agents communicating through bounded, lock-free channels. Agents process independent data blocks in parallel; backpressure and channel closure guarantee deadlock-free execution and no data loss.

Outputs

In a single pass Magic2 produces: (i) primary alignments in BAM format (or the compact `.hits` format with `--hits`); (ii) splice-junction tables with per-junction read support; (iii) gene-level expression counts with bidirectional-overlap resolution; (iv) strand-specific wiggle coverage tracks (plus read-start/end profiles for transcript boundary detection); (v) poly(A)/poly(T) tail calls and trans-spliced leader identification (in *C. elegans*); (vi) adapter consensus sequences (de-novo or from a supplied list); (vii) a standardised QC table in Tag–Sample Format (TSF) covering alignment rates, mapping-class fractions, strand-specificity, mismatch distributions, and concordant-pair statistics.

Benchmark

[PLACEHOLDER — to be completed with final numbers.]

We benchmarked Magic2 against STAR (v. X.X), HISAT2 (v. X.X), Minimap2 (v. X.X), and Magic-BLAST (v. X.X) on a panel of 35 publicly available datasets downloaded from the NCBI SRA, each comprising approximately 2 Gb of reads (Table ??). The panel includes:

- single-end and paired-end Illumina 150 nt reads (majority human);
- six cross-species datasets (chimpanzee to lemur) aligned to the human T2T reference genome, providing a controlled divergence gradient for sensitivity testing;
- approximately 14 Nanopore and PacBio long-read datasets;
- RefSeq mRNA sequences used as a zero-SNP, perfect-alignment control.

All tools were run with default parameters except where noted; adapter trimming for competing tools used Trimmomatic (v. X.X) with manufacturer-recommended settings. Magic2 required no additional configuration for any dataset.

[INSERT summary figure: heatmap or radar plot of per-tool, per-metric scores across all 35 datasets.]

[INSERT Table 2: CPU time, elapsed time, peak memory for each tool on the human paired-end reference dataset.]

Across the full panel, Magic2 ranked first or tied for first on [X/Y] quality metrics (alignment rate, concordant-pair fraction, perfect-alignment count, mismatch rate, and transcript-level precision). HISAT2 was the fastest tool in elapsed time but showed substantially lower sensitivity, particularly on divergent-species and long-read datasets. On the RefSeq control, Magic2 achieved [X] % perfect alignments, confirming correctness of the seed-sort-merge pipeline. CPU time for Magic2 was [X]-fold lower than STAR and comparable to Minimap2 on 16 threads.

Discussion

Magic2 achieves the primary goal set out in its design: a single executable that handles short and long reads, requires no configuration, and produces a richer set of outputs than any comparable tool in a single pass. The benchmark results confirm that automation of parameter selection does not come at the cost of accuracy — Magic2 matches or exceeds specialised tools on their own preferred data types.

The standardised QC output is, we argue, the feature with the broadest long-term impact. Applied at scale to the SRA, it would allow independent assessment of data quality without rerunning every published analysis, directly addressing the reproducibility concerns raised in the Introduction.

Magic2 is under active development. Planned near-term additions include GitHub Actions CI/CD for automated multi-platform builds (Linux x86-64, Linux ARM, macOS), a regression test suite, and expanded documentation.

Methods

Index construction

Given a reference genome in FASTA format and an optional GTF/GFF annotation, Magic2 builds an on-disk seed index. The seed length k (default 16 for genomes <1 Gb, 18 for human/mouse) is selected automatically from genome size. Seeds are sub-sampled with a rolling-min-hash approach (window p , default 5) to achieve an average inter-seed spacing of $p/2$ bases with a guaranteed maximum gap of p . Seeds with genomic multiplicity $> Q$ (default 81) are discarded to limit hits from repetitive elements. When annotation is provided, junction-spanning seeds from all annotated exon-exon junctions are added to the index to improve detection of reads crossing short exons. Full details are given in Supplementary Section 2.

Alignment

Read seeds are generated using the same parameters as the genome index, merged-sorted with the genomic seed table, and joined by simultaneous linear traversal. Seed matches are extended to full alignments using a banded Smith-Waterman variant; intron boundaries are refined by splice-signal scoring (GT-AG, AT-AC, and non-canonical). Poly(A)/poly(T) tails and trans-spliced leader sequences are detected during the same pass. See Supplementary Sections 3–5 for algorithmic details.

Parallelism

Magic2 spawns concurrent agents connected by bounded lock-free channels. On machines with multiple NUMA nodes, it re-spawns itself via `numactl` to bind to the least-loaded node; this step is skipped automatically in containers and on single-node machines. Users do not set a thread count; Magic2 manages concurrency automatically.

Benchmark datasets and evaluation

[PLACEHOLDER — dataset accession table and evaluation metric definitions to be inserted here.]

Code availability

Magic2 source code is available at <https://github.com/NLM-DIR/Magic2> (branch `master`) under [LICENSE]. Pre-built binaries for Linux x86-64 are attached to each GitHub Release.

Acknowledgements

[To be completed.]

Supplementary Material

See separate file `magic2_supplementary.tex`.